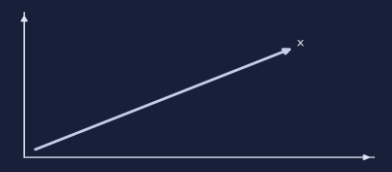
Machine Learning Foundations: Geometry, Optimization, and Linear Classification

From vectors and planes to loss functions, feature vectors, and classifiers

Dr. Yael Demedetskaya

Point / Vector

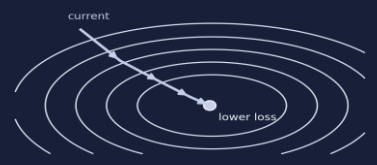


$x=(3,2,2)$

Plane / Separator

$3x_1+x_2-1=0$

Loss / Optimization



$\theta \leftarrow \theta-\gamma\nabla L(\theta)$

Classifier

$h(x)\in\{-1,+1\}$

Geometry gives us representation • Optimization gives us learning • Classification gives us prediction

Foundations for early machine learning concepts and linear classification

Part I — Framing, Notation, and Roadmap

How the course’s mathematical language becomes machine learning language

vector

hyperplane

classifier

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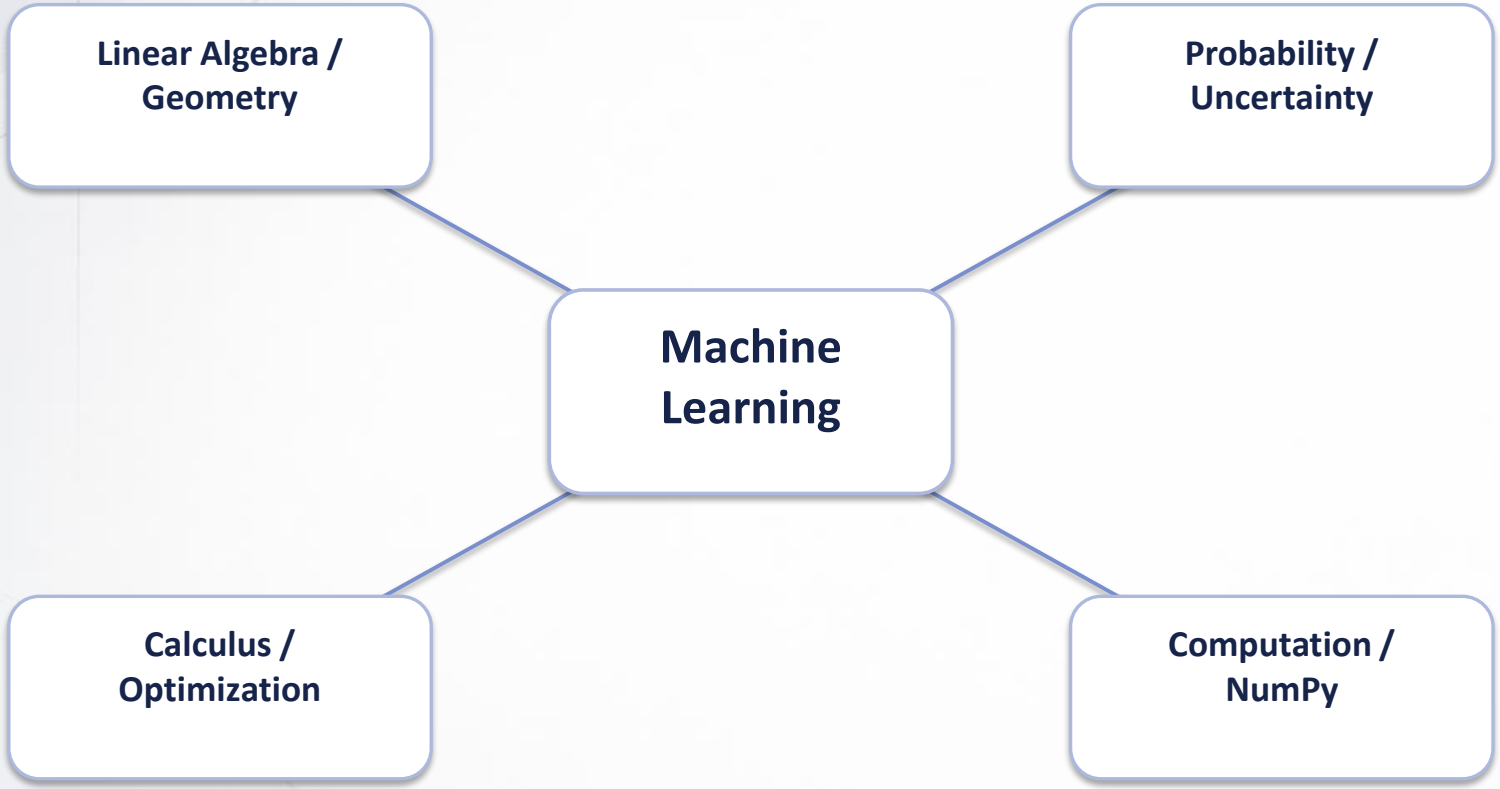
Why These Foundations Come First



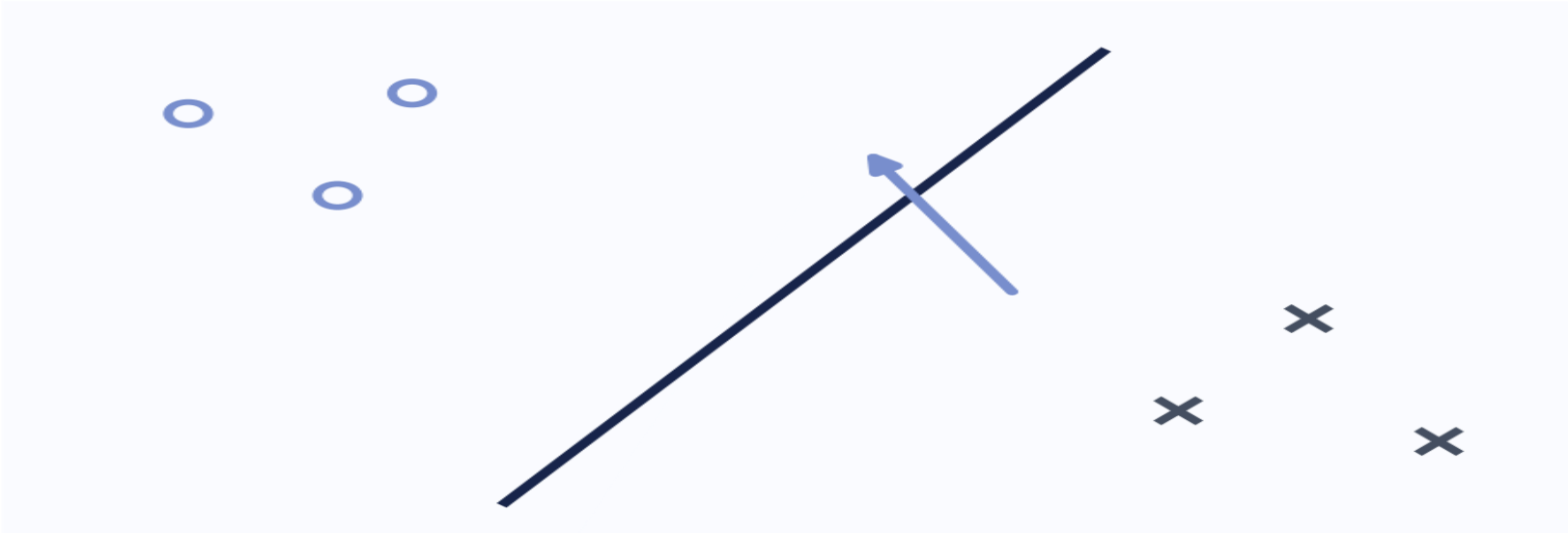
Core idea

Machine learning depends on representation, error measurement, optimization, uncertainty, and computation.

$x \mapsto$ feature vector • $L(\theta)$ • $\nabla\theta L(\theta)$

Representation, error, improvement, uncertainty, and computation all arrive before learning can happen.

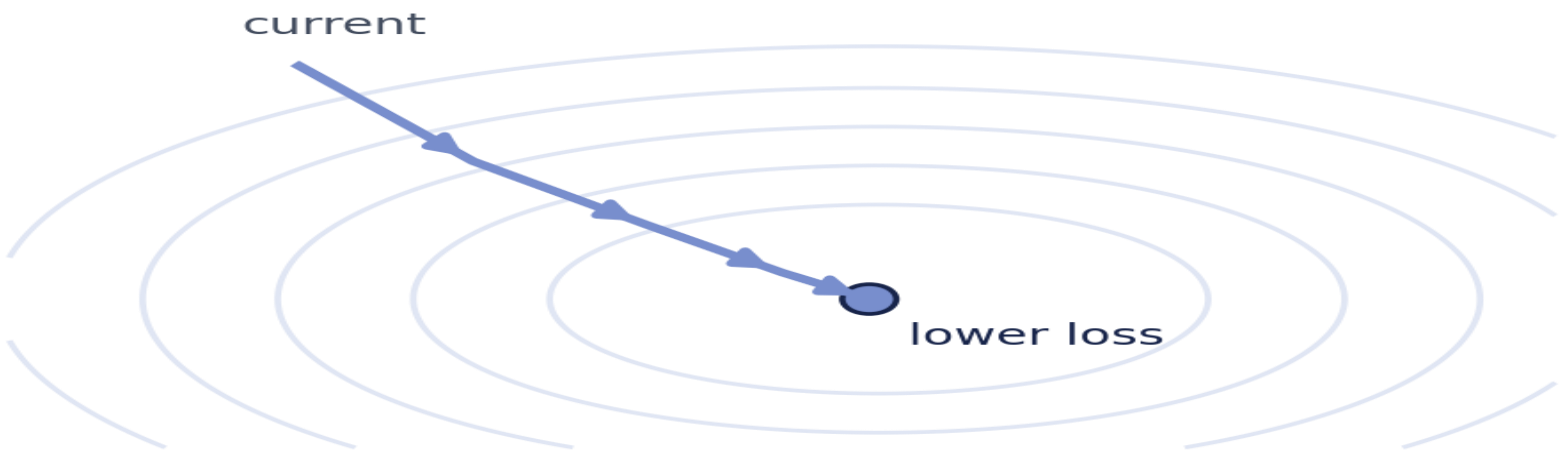
Geometry Gives Us Representation



Geometry lets us place examples in space, compare directions, and define boundaries.

Without geometry, there is no meaningful space in which learning can happen.

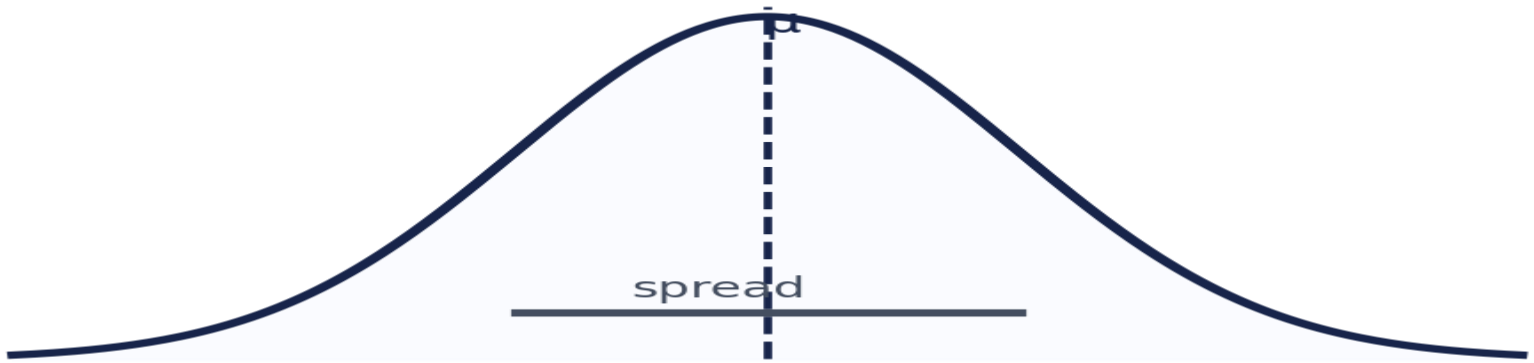
Optimization Improves Parameters



Optimization gives us a rule for improving parameters by moving toward lower loss.

Optimization is how a model changes from a bad fit into a better fit.

Probability Models Uncertainty

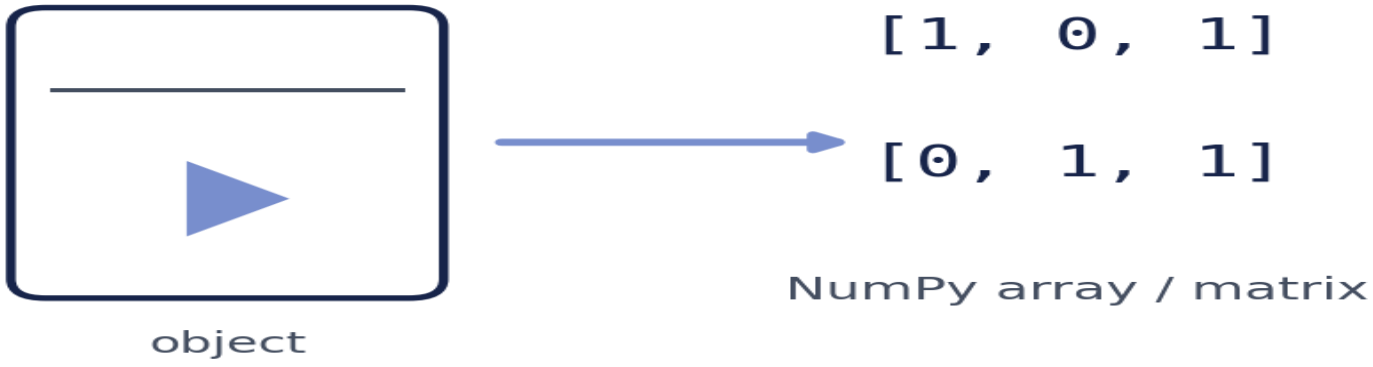


Probability lets us describe uncertainty, variability, and noisy data.

$$X \sim N(\mu, \sigma^2)$$

Real data is not perfectly clean, so machine learning must reason probabilistically.

Computation Makes the Math Executable



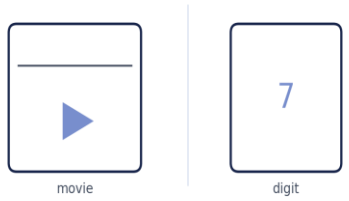
Computation turns the math into something we can evaluate, repeat, and optimize.

Without computation, the formulas stay on paper.

The Big Story: From Data to Decisions

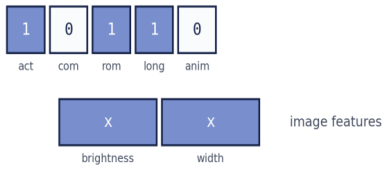
Machine learning turns real-world objects into numerical representations, places them in a geometric space, and learns rules that predict labels or values.

Raw Object



real-world input

Feature Vector

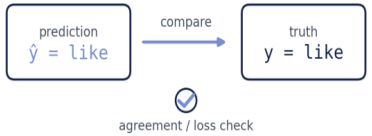


machine-readable x

Geometry / Model

space + boundary

Prediction

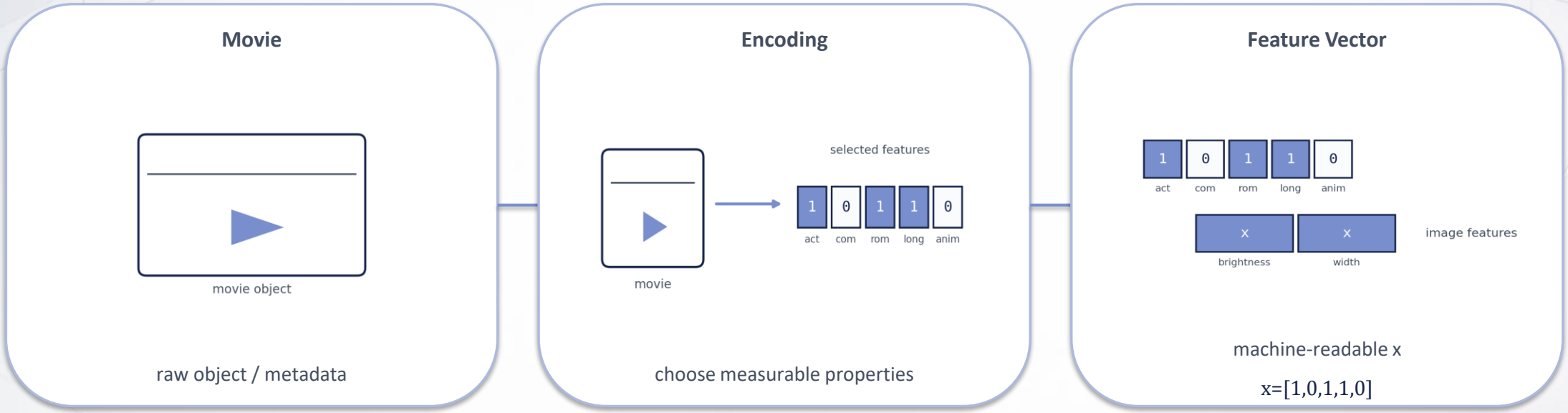


label / error

Representation comes first: objects become vectors, vectors enter geometry, and geometry supports prediction.

Example A — From Movie Object to Feature Vector

A real object becomes a machine-readable representation.

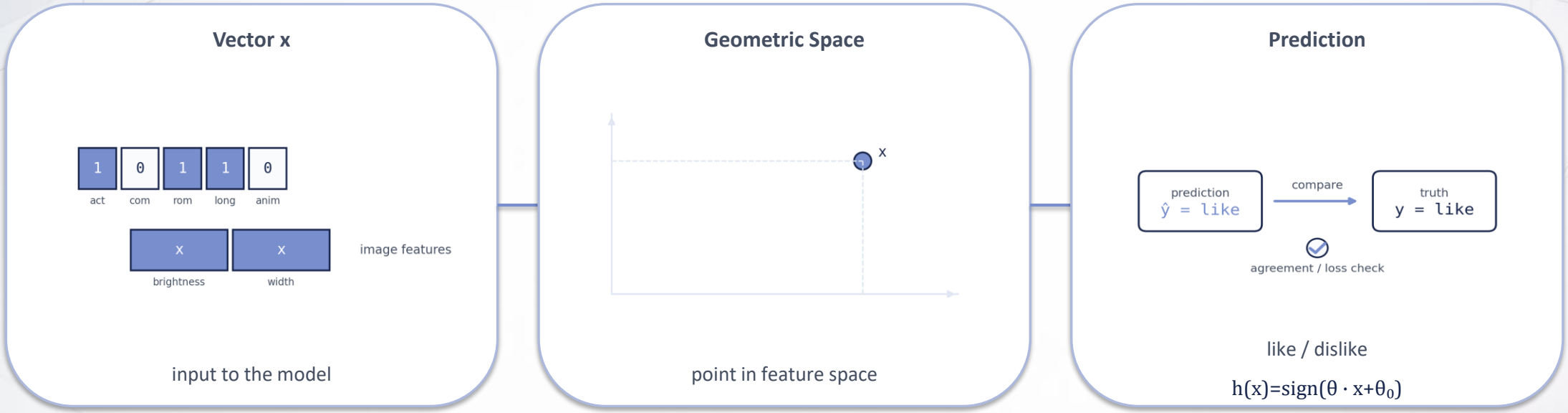


object • encoding • vector

Representation means converting a real object into a vector that a model can use.

Example A — From Feature Vector to Prediction

Once represented numerically, the object can enter a geometric model and produce a prediction.



vector • space • prediction

A prediction is made only after the object has been turned into a vector.

Example B — From Digit Image to Features

A handwritten image can also be converted into a numerical representation.

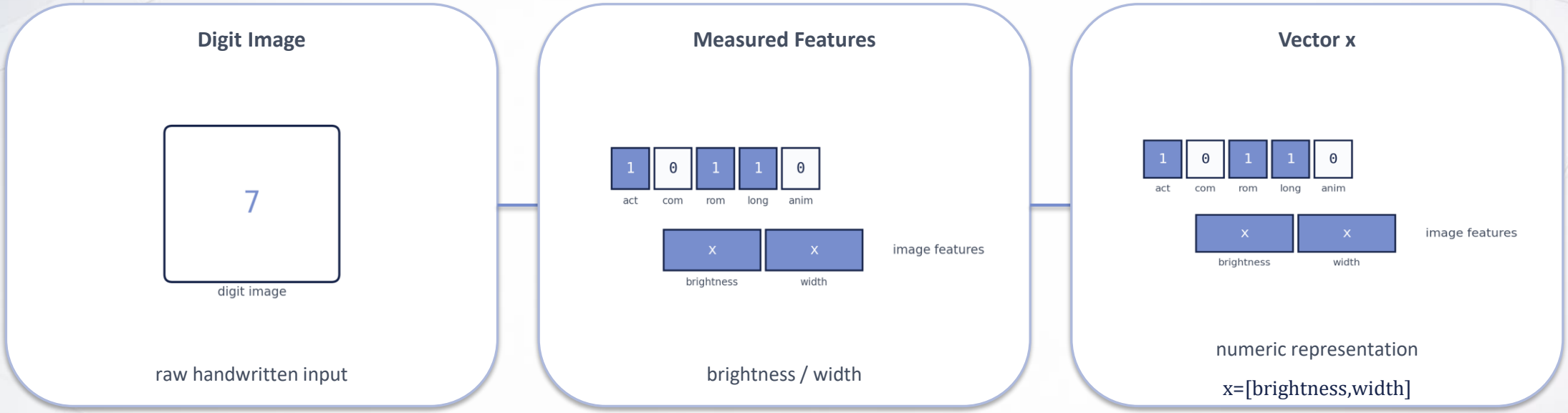
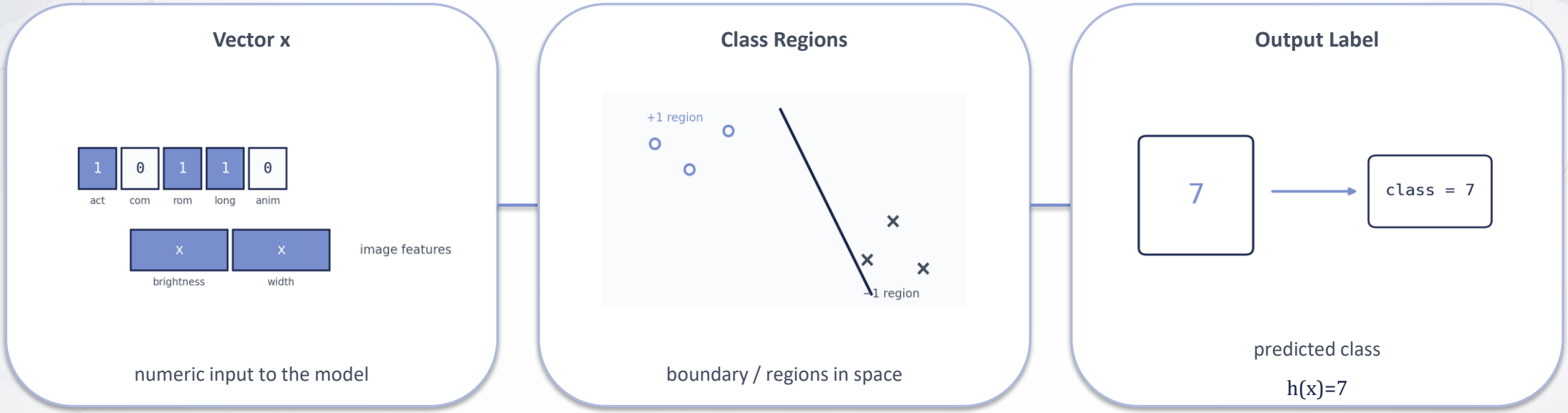


image • measurement • vector

The same representation idea works for images too.

Example B — From Features to Class Label

Once represented numerically, the image can be placed in space and classified.



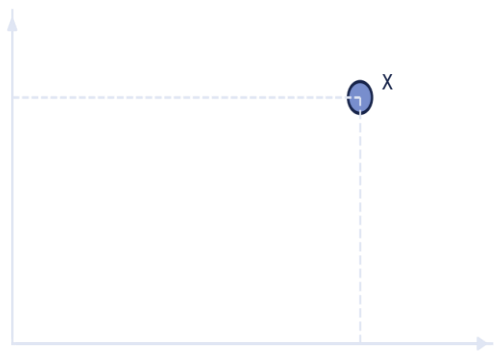
vector • boundary • class

A class prediction is made after the image becomes a vector in a geometric space.

Running Examples We Will Reuse — Geometry and Optimization

These examples will return throughout the deck as anchors.

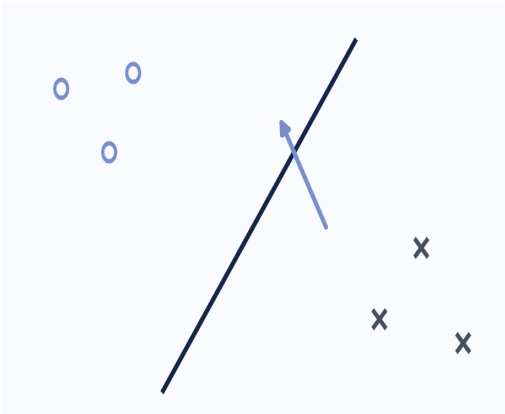
Point / Vector



$x=(3,2,2)$

geometry anchor

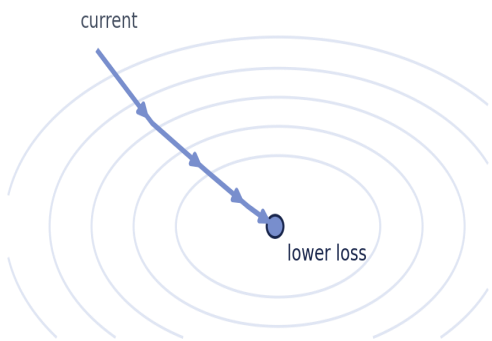
Plane / Hyperplane



$3x_1+x_2-1=0$

boundary anchor

1D Optimization



$f(x)=\frac{1}{3}x^3-x^2-3x+10$

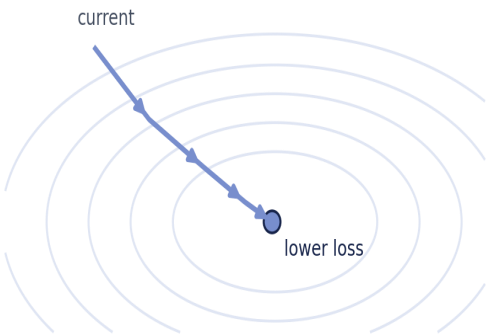
optimization anchor

geometry and optimization examples we will reuse often

Running Examples We Will Reuse — Probability, Computation, and Supervised Learning

These examples will return throughout the deck as anchors.

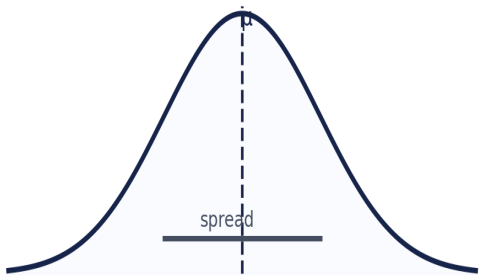
Gradient



$f(\theta)=\theta_1^2+\theta_2^2$

descent anchor

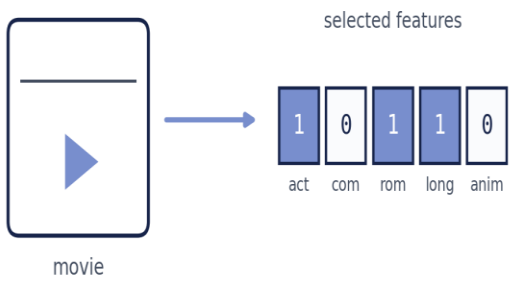
Gaussian



$X \sim N(1,2)$

probability anchor

Movie Recommender



5-feature vector + label ±1

supervised-learning anchor

probability, computation, and supervised-learning anchors

Notation You Must Read Fluently — Objects and Coordinates

In machine learning, the same letters are reused across different mathematical objects. Context matters.

symbol	meaning	visual example
$\mathbf{x}$	point / feature vector / example	$x=(3,2,2)$
$\mathbf{x_i}$	i-th coordinate	$x_2=2$
$\mathbf{A}$	matrix	A x
$\mathbf{y}$	label	$y=+1$

$x \in R^d$ • $A \in R^{m \times n}$ • $y \in \{-1,+1\}$

Notation You Must Read Fluently — Parameters, Classifiers, and Loss

These symbols will appear repeatedly in linear models, classification, and optimization.

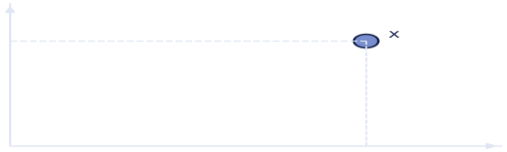
symbol	meaning	visual example
$\boldsymbol{\theta}$	parameter vector / normal vector	$\theta \cdot x + \theta_0 = 0$
$\boldsymbol{\theta_0}$	offset / intercept	boundary shift
$\mathbf{h}$	classifier / prediction rule	$h(x) = \mathrm{sign}(\theta \cdot x + \theta_0)$
$\mathbf{L}$	loss	$L(\theta)$

$\theta \in R^d$ • $\theta \cdot x + \theta_0 = 0$ • $h(x) = \mathrm{sign}(\theta \cdot x + \theta_0)$ • $L(\theta)$

Point, Vector, Feature Vector: Same Shape, Different Role

The coordinates did not change. The interpretation changed.

Point



location in coordinate space

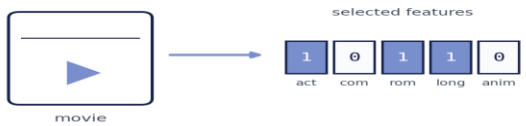
x=(3,2,2)

Vector

arrow from the origin

x∈R^3

Feature Vector



encoded example for prediction

x↦feature vector

location in space • displacement from origin • encoded example for prediction

x=(3,2,2) • x∈R^3 • x↦feature vector

same coordinates • different role • context changes meaning

Part II — Points, Vectors, Norms, Dot Products, and Projections

The geometric language used to represent examples, compare directions, and measure similarity

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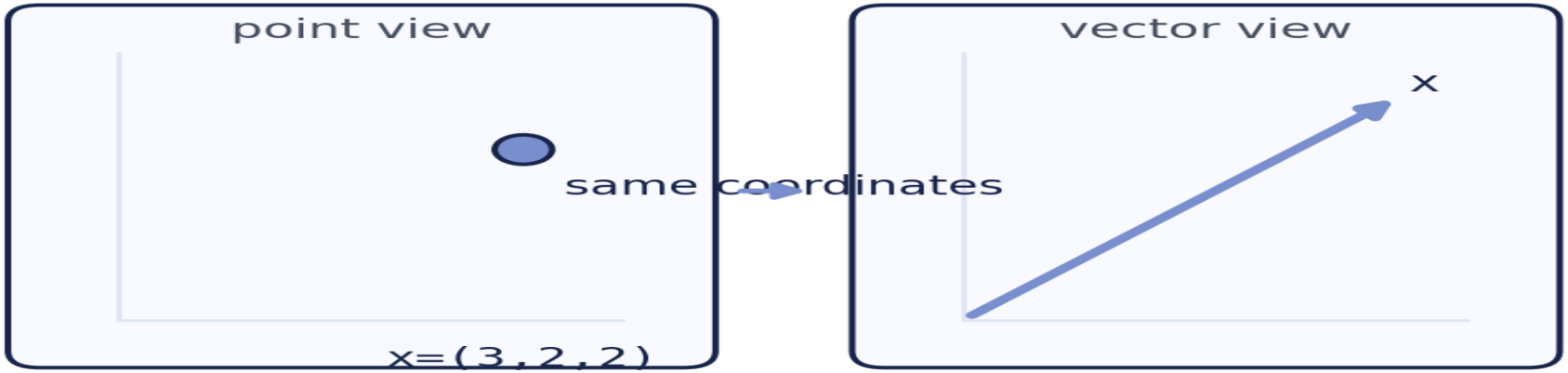
Point in Space, Vector from the Origin



The same coordinates can be interpreted in two ways: as a location in space and as a displacement from the origin.

$x=(3,2,2)$ • $x \in R^3$

This is the first bridge between geometry language and machine-learning representation.

One Set of Coordinates, Two Interpretations

Nothing numerical changed. Only the interpretation changed.

Point

location in coordinate space

$x=(3,2,2)$

Vector

displacement from the origin

$x=(3,2,2)$

Same coordinates. Different role.

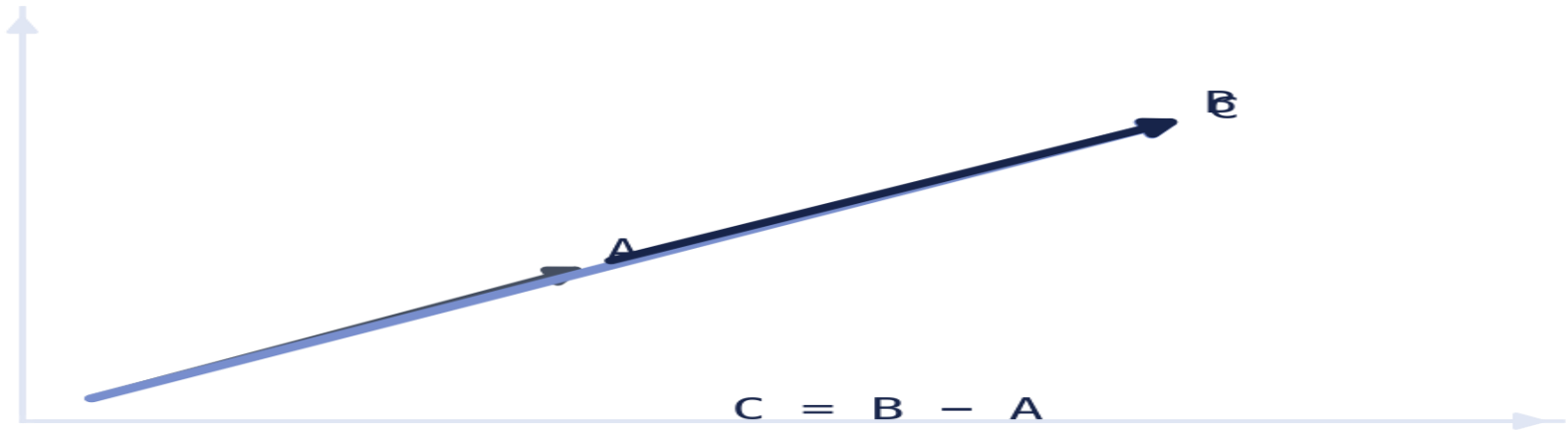
Coordinates and the Meaning of x_i

A vector is made of coordinates, and x_i means the i-th coordinate of the vector.

symbol	meaning	visual example
$\mathbf{x}$	the full vector	$x=(3,2,2)$
$[x_1,x_2,x_3]^T$	column-vector notation	$[3,2,2]^T$
x_i	the i-th coordinate	$x_2=2$
x_1,x_2,x_3	coordinate-by-coordinate view	3,2,2

$x=[x_1,x_2,x_3]^T$ • $x=(3,2,2) \Rightarrow x_1=3, x_2=2, x_3=2$

Vector Difference as Displacement

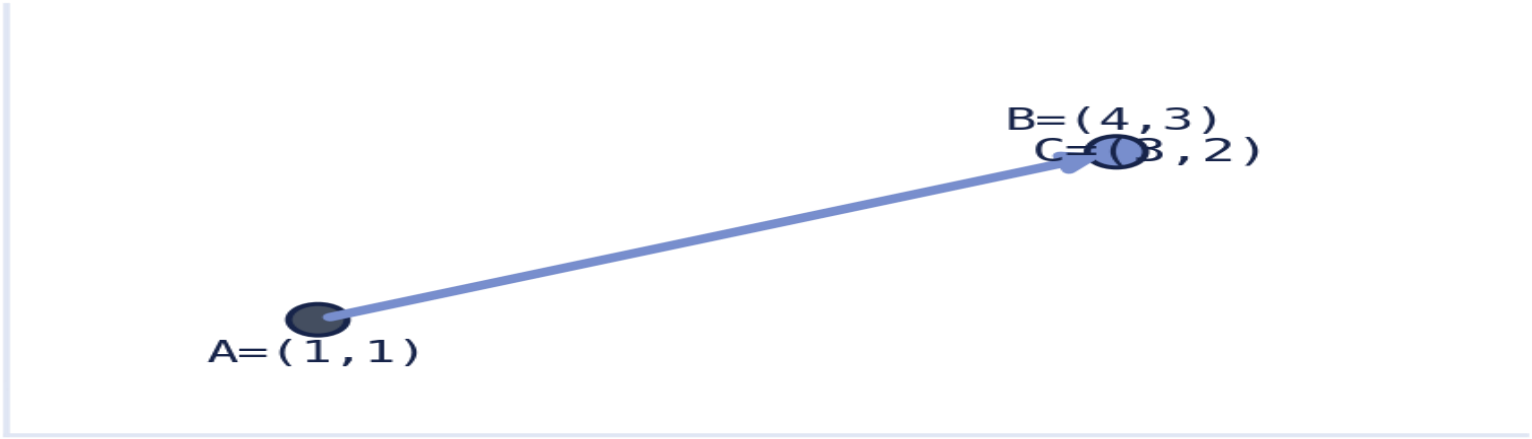


If A and B start at the origin, then the vector from A to B is their difference.

$$A + C = B \quad \bullet \quad C = B - A$$

Subtraction tells us how to move from one vector to another.

Worked Example: From One Point to Another

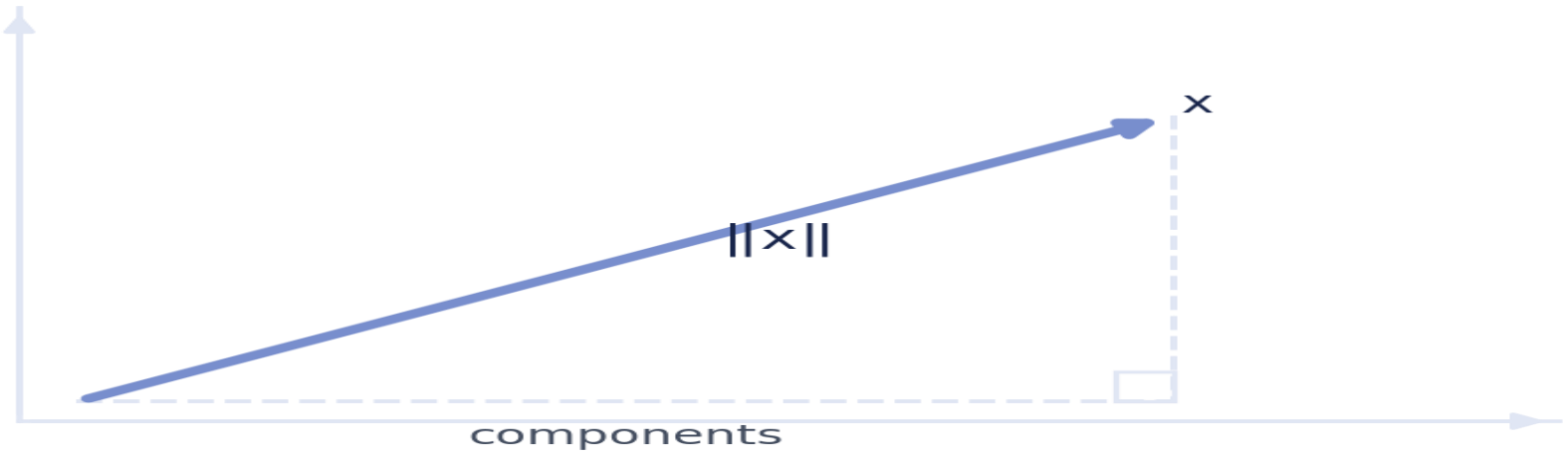


To go from A to B, subtract coordinates.

$A=(1,1)$, $B=(4,3)$ • $C = B - A$ • $C=(4,3)-(1,1)=(3,2)$

This kind of subtraction later reappears in optimization and gradients.

Norm = Length of a Vector

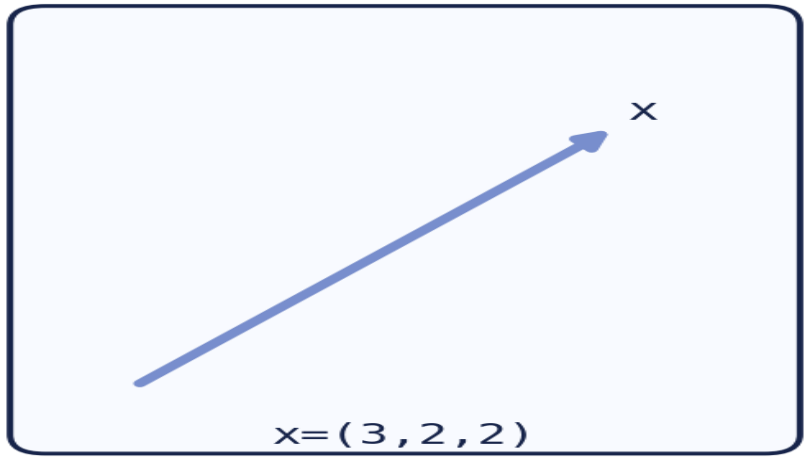


The norm tells us how long a vector is. In Euclidean space, length comes from summing squared coordinates and taking a square root.

$\|x\| = \sqrt{(x_1^2 + x_2^2 + \cdots + x_n^2)}$ • $\|x\| = \sqrt{(\Sigma_{i=1}^n x_i^2)}$

This is the first standard measurement tool for vectors.

Worked Example: ||(3,2,2)||



$$x = (3,2,2)$$

$$\|x\| = \sqrt{(3^2+2^2+2^2)}$$

$$\|x\| = \sqrt{(9+4+4)}$$

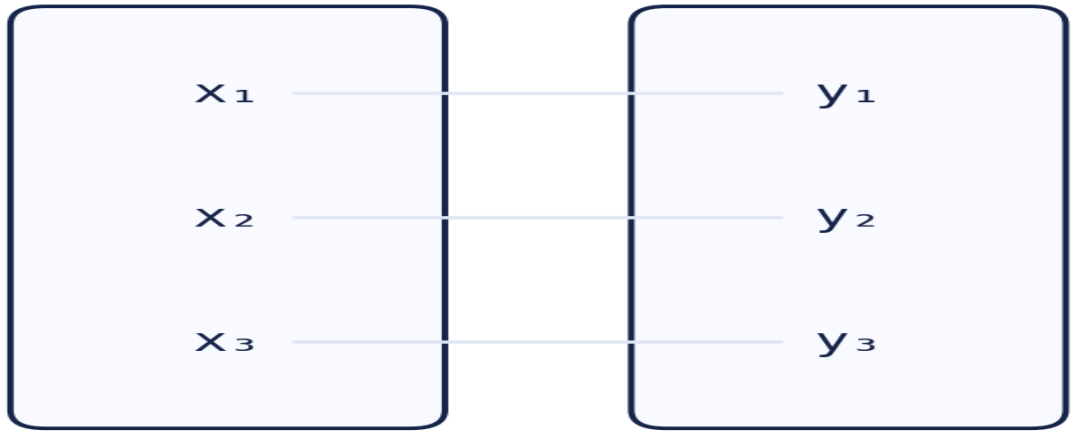
$$\|x\| = \sqrt{17}$$

Square each coordinate, add the squares, then take the square root.

x=(3,2,2) • ||x|| = √(3^2 + 2^2 + 2^2) • ||x|| = √17

Students should be able to reproduce this pattern on their own.

Dot Product as Coordinate-by-Coordinate Multiplication



$$x \cdot y = \Sigma\ x_i\ y_i$$

The dot product combines two vectors into one number. Algebraically, it is the sum of coordinate-wise products.

$x \cdot y = \Sigma_\{i=1\}^n\ x_i\ y_i$ • $x \cdot y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$

Dot product is one of the central operations in all of machine learning.

Worked Example: (3,2,2)·(1,1,1)

$$x=(3,2,2),\quad y=(1,1,1)$$

$$x\cdot y = 3\cdot 1 + 2\cdot 1 + 2\cdot 1$$

$$x\cdot y = 3 + 2 + 2$$

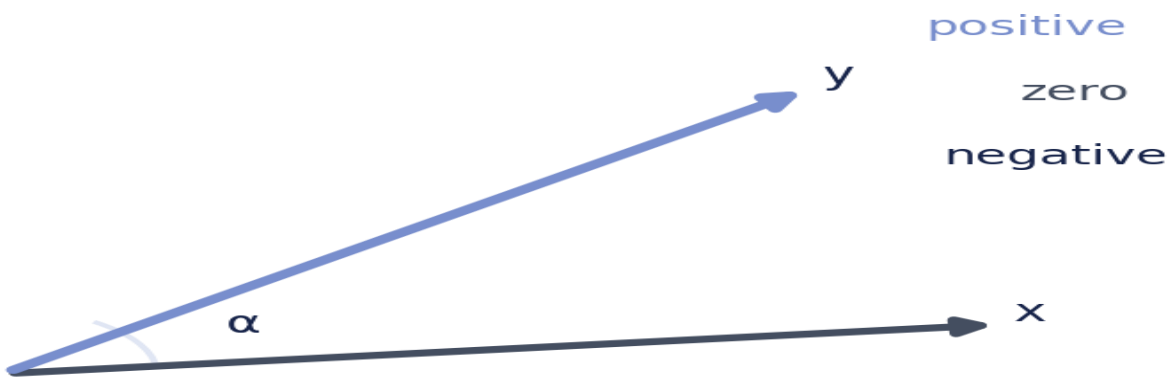
$$x\cdot y = 7$$

Dot product is often easiest to compute directly from coordinates.

$x=(3,2,2), y=(1,1,1)$ • $x\cdot y = 3\cdot 1 + 2\cdot 1 + 2\cdot 1$ • $x\cdot y = 7$

Students should be able to do this one quickly and correctly.

Dot Product as Angle and Alignment



positive

zero

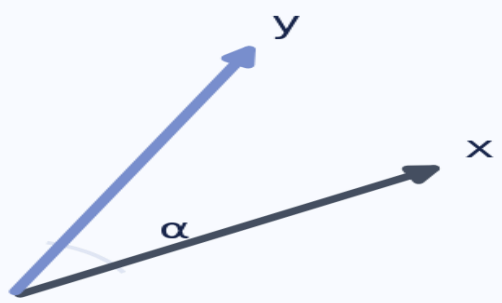
negative

Geometrically, dot product measures alignment. It depends on both length and angle.

$$x \cdot y = \|x\| \, \|y\| \cos \alpha$$

This is where the algebra starts to acquire geometric meaning.

Using the Dot Product to Find an Angle



$$\cos\alpha = (x \cdot y)/(\|x\|\|y\|)$$

$$\alpha = \arccos(\ldots)$$

If we know the dot product and the norms, we can recover the angle.

$\cos\alpha = (x \cdot y)/(\|x\|\|y\|)$ • $\alpha = \arccos((x \cdot y)/(\|x\|\|y\|))$

This is the bridge from arithmetic to geometric interpretation.

Worked Example: Norms and Angle in 2D

$$x=[0.4,0.3]^T \Rightarrow \|x\| = 0.5$$

$$y=[-0.15,0.2]^T \Rightarrow \|y\| = 0.25$$

$$x\cdot y = (0.4)(-0.15) + (0.3)(0.2) = 0$$

$$\cos\alpha = 0 \Rightarrow \alpha = \pi/2$$

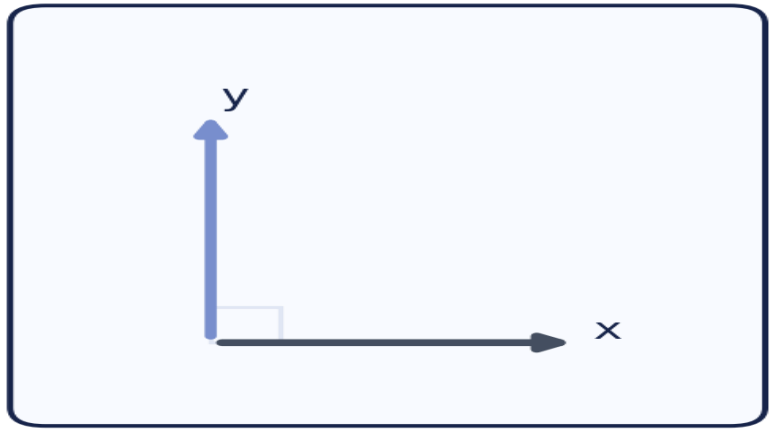
These vectors are a beautiful example because the dot product becomes zero.

compute both norms • compute the dot product • conclude the angle

$\|x\| = 0.5$, $\|y\| = 0.25$ • $x\cdot y = (0.4)(-0.15) + (0.3)(0.2) = 0$ • $\cos\alpha = 0 \Rightarrow \alpha = \pi/2$

Norm, dot product, angle, and geometry now work together.

Orthogonal Vectors



$$x \perp y \quad \square \quad x \cdot y = 0$$

$$x^{(1)} \cdot x^{(2)} = a_1^2 - a_2^2 + a_3^2$$

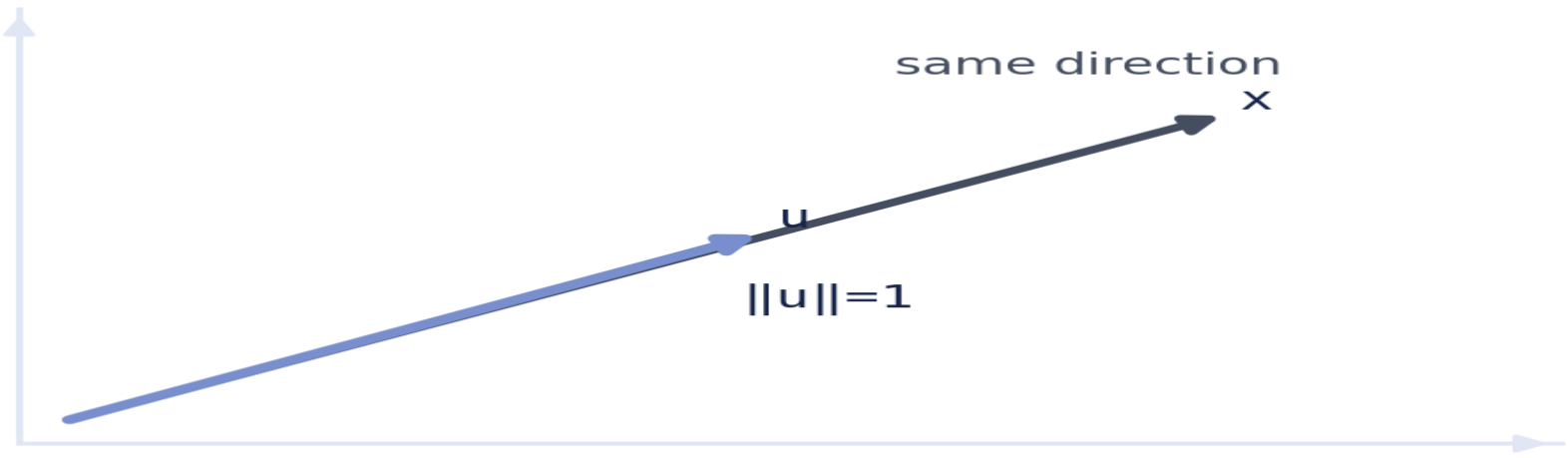
orthogonal if this equals 0

Two vectors are orthogonal when the angle between them is 90°, which means their dot product is zero.

$x \perp y \Leftrightarrow x \cdot y = 0$ • $x^{(1)} \cdot x^{(2)} = a_1^2 - a_2^2 + a_3^2$

Perpendicularity becomes a simple algebraic test.

Unit Vector = Direction Only

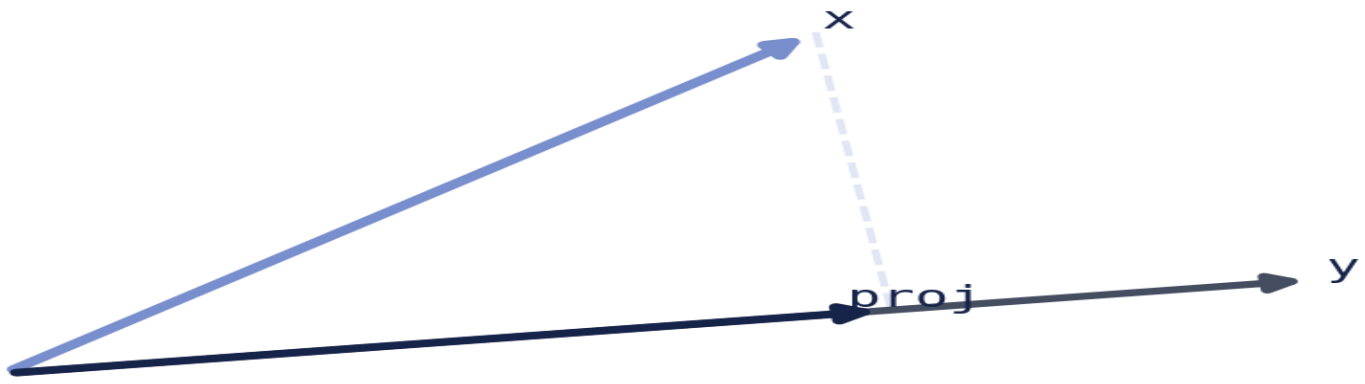


A unit vector keeps direction but removes scale.

$u = x / \|x\|$ • $x=(3,2,2) \Rightarrow u=(1/\sqrt{17})(3,2,2)$

This is how we isolate pure direction.

Projection = How Much of x Lies in the Direction of y

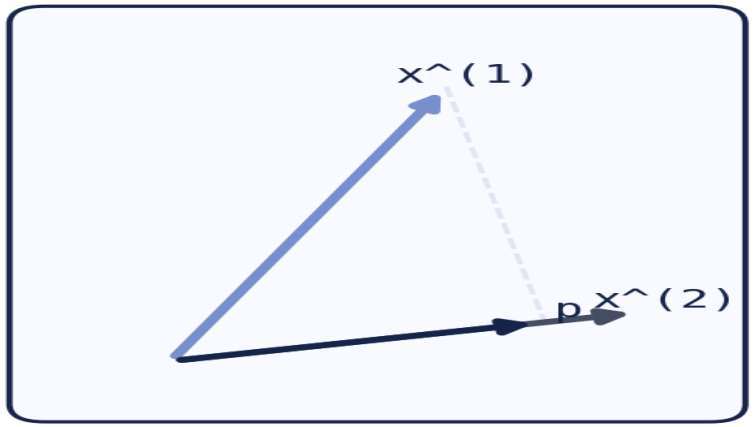


Projection asks what part of one vector points along another. It isolates the directional component.

$\text{proj_y}(x) = ((x \cdot y)/(\|y\|^2))\, y$ • if $u = y/\|y\|$, then $\text{proj_y}(x) = (x \cdot u)u$

Projection is one of the most useful geometric decompositions in the deck.

Projection of x^(1) Onto x^(2)



$$u = x^{(2)}/\|x^{(2)}\|$$

$$p = c\ u$$

$$c = (x^{(1)} \cdot x^{(2)})/\|x^{(2)}\|$$

$$x^{(1)} \cdot x^{(2)} = a_1^2 - a_2^2 + a_3^2$$

The projection must point in the direction of x^(2). If u is the unit direction, then projection can be written as c u.

$u = x^{(2)}/\|x^{(2)}\|$ • $p_{x^{(1)} \to x^{(2)}} = c\ u$ • $c = (x^{(1)} \cdot x^{(2)})/\|x^{(2)}\|$ • $x^{(1)} \cdot x^{(2)} = a_1^2 - a_2^2 + a_3^2$

This is the projection question in a form students can actually follow and reproduce.

Why Norms, Dot Products, and Projections Matter in ML

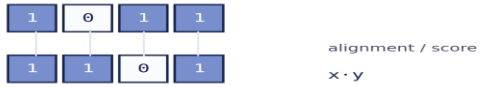
Norms measure size, dot products measure alignment, and projections isolate directional components.

Norm

feature-vector size / parameter magnitude

$\|x\|$

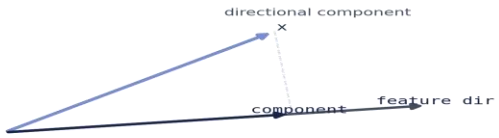
Dot Product



similarity / alignment / scoring

$x \cdot y$

Projection



directional component / useful subspace

$\mathrm{proj}_y(x)$

norm → size / magnitude • dot product → similarity / alignment • projection → directional component

This geometry is the operating language of machine learning.